ICT 3144 Embedded System Lab Synopsis , JUL-NOV 2026

Vth SEM B.Tech. (IT)

Date of submission: 8th oct ,2026

PROJECT TITLE: LPC1768-Based Interactive Quiz and Score System

BRIEF DESCRIPTION ABOUT THE PROJECT:

The proposed project is an **LPC1768-Based Interactive Quiz and Score System** designed using the LPC1768 microcontroller. The system allows a user to participate in a multiple-choice quiz by selecting answers through a keypad. The questions, options, and correct answers are stored in the microcontroller program, eliminating the need for any external storage or additional modules.

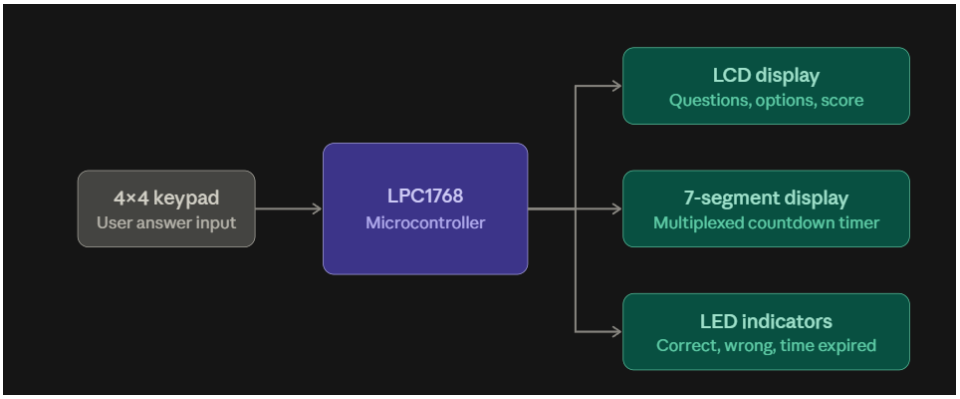
At the beginning of the quiz, the LCD displays a welcome message and prompts the user to start the quiz. Each question and its multiple-choice options are displayed on the LCD. Since a standard 16×2 LCD has limited display space, the question and options can be displayed across multiple screens. The user selects an option using the keypad, and the LPC1768 compares the selected answer with the correct answer stored in the program. The score is then updated accordingly.

A multiplexed seven-segment display is used to display a countdown timer for each question. For example, the user may be given 10 seconds to answer each question. If the time expires, the system automatically considers the question unanswered and proceeds to the next question. LEDs are used to indicate the result of an answer, such as a green LED for a correct answer and a red LED for an incorrect answer.

After all questions have been attempted, the final score is calculated and displayed on the LCD. The user can then restart the quiz using the keypad. The project demonstrates important embedded-system concepts such as **GPIO interfacing, keypad scanning, LCD interfacing, multiplexed seven-segment display control, timer-based operations, LED control, conditional logic, and state-based program execution** using the LPC1768 microcontroller.

The system is designed to be simple, low-cost, and suitable for an Embedded System Laboratory while demonstrating the practical use of multiple peripherals of the LPC1768.

METHODOLOGY / PORT LINES USED AND CONNECTIONS



Basic Working Flow:

Start → Display Welcome Screen → Press Start → Display Question → Display Options → Read Keypad Input → Check Answer → Update Score → Display Result → Next Question → Display Final Score → Restart

Port Lines and Connections:-

Peripheral Module	FRC Connector	Port Lines / Pins Used	Functional Description
MSSD Data	**CNA**	P0.4 – P0.11 (Pins 1-8)	Segment Data Lines
MSSD Select	**CNC**	P0.15 – P0.18 (Pins 1-4)	Decoder Digit Select Lines
4x4 Keypad	**CNB**	P1.23 – P1.26 (Cols, Pins 1-4) P2.10 – P2.13 (Rows, Pins 5-8)	Matrix Keypad Scanning
16x2 LCD	**CND**	P0.23 – P0.26 (Pins 1-4) P0.27 – P0.28 (Control, Pins 5-6)	4-Bit LCD Control & Data

COMPONENTS REQUIRED

1. LPC1768 Microcontroller/Development Board
2. 16×2 LCD Display
3. 4×4 Matrix Keypad
4. Multiplexed Seven-Segment Display
5. LEDs
6. Jumper Wires
7. Power Supply/USB Cable

TEAM MEMBERS

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